TRENESIS

# STRENGTH TRAINING GUIDE

Evidence-based fundamentals for building real, lasting strength.

| 5 | 3 | 30 |
|---|---|---|
| CORE PRINCIPLES | FULL PROGRAMS | EXERCISES |

NO FLUFF · NO PSEUDOSCIENCE · JUST RESULTS

BEFORE YOU START

# Disclaimer & How to Use This Guide

This guide is for informational and educational purposes only. It does not constitute medical advice. Consult a qualified healthcare professional before starting any new exercise program, especially if you have pre-existing injuries or health conditions.

The information presented is based on current evidence-based exercise science. Individual results vary based on consistency, nutrition, sleep, and genetics.

## How to Use This Guide

- Read Chapters 1–3 before picking a program. The principles make everything else click.
- Choose the program in Chapter 4 that fits your schedule and current level.
- Log every session — sets, reps, weight. Progress requires data.
- Come back to Chapter 5 when you plateau. Most stalls are lifestyle issues.

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CHAPTER 01

# Why Strength Training?

The science-backed case for lifting heavy.

## The Case for Getting Strong

Strength training is the single most impactful thing most people can do for their long-term health and body composition. The evidence is unambiguous: resistance training improves bone density, increases resting metabolic rate, enhances insulin sensitivity, reduces injury risk, and is directly linked to lower all-cause mortality.

Yet most people who join a gym waste years doing ineffective programs. This guide fixes that. You will learn the fundamentals that produce results across all goals — whether you want to look better, perform better, or simply live healthier.

## Key Benefits

| | |
|---|---|
| **Muscle Mass** | Increases resting metabolic rate. More muscle = more calories burned 24/7. |
| **Bone Density** | Mechanical loading stimulates bone mineralisation, reducing osteoporosis risk. |
| **Hormonal Health** | Optimises testosterone, growth hormone, and insulin sensitivity in both sexes. |
| **Mental Health** | Consistent resistance training significantly reduces depression and anxiety. |
| **Injury Prevention** | Stronger muscles, tendons and connective tissue protect joints under load. |
| **Longevity** | Grip strength and lower-body power are among the best predictors of healthy aging. |

> **KEY INSIGHT**
>
> Every kilogram of muscle you add burns roughly 13 extra kcal per day at rest. That compounds over time — and it works even while you sleep.

CHAPTER 02

# The 5 Core Principles

Master these and everything else becomes detail.

## 01 Progressive Overload

The foundational rule of all strength training. Your body adapts to the demands placed on it — to keep improving, those demands must increase over time. This does not always mean adding weight. You can progress by adding reps as well. But you must progress. Track every session. Side note: This doesn't mean you should sacrifice your technique to force progressive overload. Progressive overload is the result of building muscle, not the other way around.

## 02 Specificity

You get better at exactly what you train. Want to squat more? Squat. Want bigger arms? Train biceps and triceps with sufficient volume and intensity. Your program should directly reflect your goals. Generic work suits beginners; as you advance, specificity becomes non-negotiable.

## 03 Recovery

Strength is not built in the gym. It is built while you rest. Training is the stimulus; recovery is when adaptation occurs. Under-recovery is the most common reason people stall. 7–9 hours of sleep, sufficient protein, and stress management are part of the program — not optional extras.

## 04 Consistency

Three mediocre sessions per week, every week, for a year beats three perfect sessions followed by two weeks off. Consistency compounds. The best program is the one you will actually follow. Do not let perfect be the enemy of good.

## 05 Technique First

Poor technique limits how much weight you can safely lift, increases injury risk, and reduces muscle activation. Before adding load, own the movement. Spend your first 4–8 weeks learning fundamental patterns at sub-maximal loads. This investment pays dividends for years.

> **THE HIERARCHY**
>
> Consistency > Technique > Progressive Overload > Program Design > Supplements. Most people obsess over the bottom of this list while neglecting the top.

CHAPTER 03

# Programming 101

How to structure your training intelligently.

## Key Training Variables

Every program is built from a small number of variables. Understanding them lets you evaluate any program — and adjust it intelligently when life gets in the way.

| VARIABLE | DEFINITION | VALUE |
|---|---|---|
| Frequency | Times per muscle group per week | 2–3 × |
| Volume | Total sets per muscle per week | 10–15 sets |
| Intensity | Degree of effort in a set | Close to muscle failure |
| Rep Range | Reps per set | 5–12 |
| Rest | Between sets | At least 2 min |

### REP RANGE

Anything between about 5–30 reps per set has a similar muscle-building stimulus if sets are taken close to failure. Since lower reps usually cause less fatigue, it's often best to do as few reps as possible, but as many as necessary for you to feel comfortable.

### PROGRESSION RULE

Once you can perform a certain number of reps with a weight (for example 12 reps), increase the weight. Then build the reps back up with the new weight and repeat the process over time.

CHAPTER 04

# The Programs

Three complete programs. Pick one and commit.

Each program is evidence-based, time-efficient, and built around the principles in Chapter 2. Pick based on your schedule and current level.

program A

## Full Body Split

A Full Body Split is a workout routine where every training session targets all major muscle groups. Instead of splitting workouts by body parts, you train the whole body about three times per week with rest days in between. It's ideal for people who can only go to the gym three times (or even two times) a week.

When planning a Full Body Split, choose one exercise for each muscle group. Use the same selection in every session, and perform 2–3 sets of each exercise.

program B

## Upper / Lower Split

Four days per week — two upper-body, two lower-body sessions. In an Upper/Lower Split, workouts are divided between upper-body muscles (chest, back, shoulders, arms) and lower-body muscles (quads, hamstrings, glutes, calves). This structure allows shorter training sessions while still giving each muscle group enough focus and recovery.

When planning an Upper/Lower Split, choose one exercise for each muscle group in that session. For upper-body days, pick exercises for chest, back, shoulders, biceps, and triceps. For lower-body days, pick exercises for quads, hamstrings, glutes, and calves. Perform 2–3 sets of each exercise, alternating between upper and lower days.

program C

## Torso / Limbs Split

Four days per week — two torso-focused, two limb-focused sessions. In a Torso/Limbs Split, workouts are divided between core and upper-back muscles (torso) and arms and legs (limbs). This setup is especially effective for emphasizing arm development, since you can place arm training at the start of limb-focused sessions when your energy is highest, allowing for more intensity.

When planning a Torso/Limbs Split, choose one exercise for each muscle group in that session. For torso-focused days, pick exercises for chest, back, shoulders, and core. For limb-focused days, pick exercises for biceps, triceps, quads, hamstrings, glutes, and calves. Perform 2–3 sets of each exercise, alternating between torso and limb days.

## Exercises

| MUSCLE | EXERCISE | NOTES |
|---|---|---|
| **CHEST** | | |
| | Bench Press | Always use a spotter to ensure safety. |
| | Dumbbell Bench Press | Not ideal for beginners because of instability. |
| | Machine Chest Press | |
| | Flyes (Dumbbell, cable or machine) | Perfect to isolate the chest. |
| **BACK** | | |
| **Lats** | | |
| | Lat Pulldown (Wide- or close-grip) | |
| | Close-grip Cable Row | Make sure to pull toward the belly button and, not toward the chest. |
| **Upper Back** | | |
| | Wide-grip Cable Row | Pull toward your chest and squeeze your shoulder blades at the end of the movement. |
| | T-Bar Row | Pull toward your chest and squeeze your shoulder blades at the end of the movement. |
| | Kelso Shrugs | Squeeze your shoulder blades (good choice if you feel your biceps too much when doing rows) |
| **Lower Back** | | |
| | Back Extension | |
| **ARMS** | | |
| **Biceps** | | |
| | Dumbbell Curls | Good choice if you only have dumbbells. |
| | Preacher Curls | Optimal for stability. |
| | Hammer Curls | Some more brachialis activation. |
| **Triceps** | | |
| | Cable Triceps Pushdown | |
| | Overhead Extension | |
| **Forearms** | | |
| | Reverse Curls | Emphasizes the brachioradialis. |
| | Wrist Curls | |
| | Wrist Extensions | |
| **SHOULDERS** | | |

| | | |
|---|---|---|
| | Lateral Raises (Cable, dumbbell or machine) | Mainly works the side delts. |
| | Shoulder Press | |
| | Reverse Flyes | Great for building rear delts. |
| **ABS** | | |
| | Crunches (Machine, cable or bodyweight) | Make sure to actually round the spine instead of just flexing your hips. |
| | Leg Raises | Make sure to actually round the spine instead of just flexing your hips. |
| **LEGS** | | |
| **Quads** | | |
| | Squats | |
| | Leg Press | Put your feet lower to emphasize the quads. |
| | Leg Extension | Important to target the rectus femoris (not activated sufficiently with squats and leg press) |
| **Hamstrings** | | |
| | Leg Curl | |
| **Glutes** | | |
| | Hip Thrusts | Great for isolating the glutes. |
| | Leg Press | Put your feet higher to emphasize the glutes. |
| **Calves** | | |
| | Standing Calf Raise | Likely to be better for calf growth than seated calf raises. |
| | Seated Calf Raise | |

CHAPTER 05

# Recovery & Sleep

Where strength gains are actually made.

## Sleep: The Most Anabolic Tool You Have

Sleep is the most anabolic state the human body can be in. During deep sleep, growth hormone is released, muscle protein synthesis peaks, and the central nervous system recovers from training stress. Chronic sleep restriction (under 6 hours) reduces testosterone, impairs recovery, increases injury risk, and tanks motivation.

## Sleep Optimisation Checklist

- Aim for 7–9 hours. Prioritise more on heavy training days.
- Maintain a consistent sleep and wake time.
- Avoid screens 30–60 minutes before bed (blue light suppresses melatonin).
- Keep your bedroom cool — 17–19 °C is optimal for sleep quality.
- Do not train within 2 hours of bed if you have trouble falling asleep.
- Avoid alcohol — it fragments sleep architecture even in small amounts.

## Active Recovery

Rest days do not have to mean complete inactivity. Light movement — walking, cycling, swimming — increases blood flow to muscles and accelerates recovery. Aim for 20–40 minutes of easy activity on rest days. Avoid anything that causes significant muscle damage or fatigue.

> **STRESS MATTERS**
>
> Psychological stress raises cortisol, which directly competes with testosterone and impairs muscle protein synthesis. Managing life stress is a legitimate training variable. Chronically stressed athletes recover slower, plateau earlier, and get injured more.

CHAPTER 06

# Common Mistakes

The errors that stall 90% of lifters.

## Not tracking progress

If you don't write it down, you won't know if you're improving. Log every set, rep, and weight. Review weekly. Progress is not obvious day-to-day — the data reveals the trend.

## Ego lifting

Using more weight than you can control reduces muscle activation, increases injury risk, and trains your ego rather than your muscles. A controlled 80 kg squat is worth more than a sloppy 120 kg quarter-squat.

## Program hopping

Switching programs every 2–3 weeks is one of the biggest mistakes in the gym. Adaptation takes time. Your muscles need time to learn new movements, and constant changes slow your progress. Real results come from sticking to a solid program and executing it consistently.

## Neglecting the posterior chain

Most people over-train pushing movements and under-train pulling movements, creating imbalances that cause pain and limit performance. Match every push set with an equal volume of pulling.

## Skipping warm-ups

A proper warm-up means 2–3 progressive warm-up sets at lighter loads before your working sets. This primes the nervous system, lubricates joints, and improves performance. Five minutes on the treadmill is not a warm-up.

## Skipping cardio

Cardio isn’t just for fat loss — it improves heart health, recovery, and work capacity in the gym. Neglecting it can limit overall fitness and endurance, even if your lifting routine is solid.

CHAPTER 07

# Frequently Asked Questions

### How long until I see results?

Strength improvements begin within 2–4 weeks (largely neurological). Visible muscle changes typically require 8–12 weeks of consistent training and adequate nutrition. Body composition changes can take 3–6 months to become obvious to others. Patience is not optional.

### Should I train when sore?

Training a sore muscle is fine if it’s light soreness. If it’s severe enough to change your form or limit your lifts, just wait one more day. Never hit the same muscle two days in a row.

### Can I build muscle while losing weight?

Yes — especially as a beginner, returning lifter, or if carrying excess body fat. Advanced lifters will find it very slow.

### Free weights or machines?

Both are effective. Free weights usually offer more versatility, but machines provide extra stability, which can be especially helpful for beginners. Ultimately, it comes down to personal preference.

### How do I know if I'm training hard enough?

A good way to gauge intensity is by using Reps in Reserve (RIR) — basically, how many more reps you could do before failure. In your first sets, aim for about 0–1 RIR, meaning you could maybe squeeze out one more rep. Pushing to failure is totally fine, especially on the last set for each muscle

### Is it bad if I’m not sore after a workout?

Not feeling sore after a workout is totally normal. Muscle soreness isn’t a reliable indicator of growth—what matters is the stimulus you give the muscle. Soreness (or having a pump) can help confirm you hit the right area, but not having them doesn’t mean your session wasn’t effective.

## Ready to put it into practice?

More free guides, training content, and resources at:

**trenesisgym.github.io/trenesis/**

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